

Users > moshika > Desktop > lab > 🐙 forward_chain.pl

```
1  /* Knowledge Base */
2  fact(ram_is_student).
3  fact(ram_has_good_marks).
4  fact(ram_is_from_poor_family).
5
6  /* Rules for Forward Chaining */
7
8  /* Rule 1: If a person is a student and has good marks, he is intelligent */
9  rule(intelligent(X)) :-
10     fact(X_is_student),
11     fact(X_has_good_marks).
12
13  /* Rule 2: If a person is intelligent and from a poor family, he gets scholarship */
14  rule(scholarship(X)) :-
15     rule(intelligent(X)),
16     fact(X_is_from_poor_family).
17
18  /* Forward Chaining Engine */
19  forward_chaining :-
20     ( rule(scholarship(ram)) ->
21         write('Ram qualifies for scholarship'), nl
22     ; write('Ram does not qualify')
23     ).
24
```

```
[moshika@Moshikas-MacBook-Air ~ % cd /Users/moshika/Desktop/lab
[moshika@Moshikas-MacBook-Air lab % swipl forward_chain.pl
Warning: /Users/moshika/Desktop/lab/forward_chain.pl:9:
Warning:     Singleton variables: [X,X_is_student,X_has_good_marks]
Warning: /Users/moshika/Desktop/lab/forward_chain.pl:14:
Warning:     Singleton variables: [X_is_from_poor_family]
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.
```

```
For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
[?- forward_chaining.
Ram qualifies for scholarship
true.
```

```
[?- rule(intelligent(ram)).
true .
```

```
[?- halt.
```